Homework 07 - Spectral density estimation & the Laplace transform

Deadline: 3rd test date (14th of May)
The mandatory questions account for 100 points, while the optional questions together contribute up to an additional 50 points.

1. Demonstrate the validity of the Parseval theorem using FFT.
2. Define the periodogram. Why is it a poor estimator?
3. What are the benefits and side effects of slicing the signal when calculating the periodogram?
4. Explain the bias–variance trade-off in spectral estimation.
5. Describe the Welch method.
6. Why does Welch assume stationarity?
7. Why does Welch reduce variance?
8. What is lost when using Welch instead of a single FFT?
9. What is the effect of segment length?
10. Why does averaging reduce variance but not bias?
11. Why is overlap used in Welch?
12. Why are windows applied before the FFT?
13. What is detrending and why is it used?
14. What is zero padding? What does it change?
15. If segments have 300 samples, what FFT length is typically used and why?
16. In spectral estimation (e.g., periodogram/Welch), detrending, windowing, and zero-padding are commonly applied before the FFT. To what extent can the order of these operations be interchanged without affecting the result? Provide explanation, identifying which operations commute, which do not, and why.
17. Why must sampling frequency appear in PSD scaling?
18. What is meant by a one-sided spectrum? When is it a valid measure of the energy content of a signal?
19. Explain the line below when normalizing a one-sided spectrum:

    `FFTSlices[:, 1:-1] *= np.sqrt(2)`
20. Implement your own Welch method and compare it to `scipy.signal.welch` using a generated harmonic signal with added noise.
21. In the context of the Welch method demonstrate numerically the effect of
    (a) segment size
    (b) overlap
    (c) detrending
    (d) window type
    (e) padding

    (for the same signal apply different parameters and briefly discuss the observed differences in the obtained PSD)
22. Read in and plot the EEG signal from the `eeg.bin` file (it is in a contiguously laid out binary format).

    Hint: amplitude is typically within 100 μV. Use `numpy.fromfile`
23. For the above signal calculate and plot the signal's power spectral density using 4 seconds long, 50% overlapping Hamming windows, with detrending.
24. Determine the above signal's sampling frequency.

    Hint: A 50 Hz notch filter removed the pollution from the power line.
25. Explain the Laplace transform and its kernel. Why is it widely used in mathematics, physics, and engineering?
26. Describe the linearity property of the Laplace transform. Provide an example demonstrating this property.
27. Prove the following properties of the Laplace transform:
    (a) $L[1] = \frac{1}{s}$, $s > 0$
    (b) $L[e^{at}] = \frac{1}{s-a}$, $s > 0$
    (c) $L[\cos(at)] = \frac{s}{s^2+a^2}$
    (d) $L[\sin(at)] = \frac{a}{s^2+a^2}$
    (e) $L[t^p] = \frac{\Gamma(p+1)}{s^{p+1}}$, $s > 0$
    (f) $L[e^{ct}f(t)] = F(s-c)$
    (g) $L[f(ct)] = \frac{1}{c}F\left(\frac{s}{c}\right)$
    (h) $L[\delta(t-t_0)] = 1, t_0 > 0,$
    (i) $L[H(t-t_0)] = \frac{e^{-st_0}}{s}, t_0 > 0$
28. Prove the convolution theorem in the context of the Laplace transform.
29. Determine the inverse Laplace transform of the functions below:
    (a) $\dfrac{3}{s^2+4}$
    (b) $\dfrac{3s}{s^2-s-6}$
    (c) $\dfrac{5s+2}{s^2+2s+5}$
    (d) $\dfrac{2s-3}{s^2-4}$
    (e) $\dfrac{8s^2-4s+12}{s(s^2+4)}$
    (f) $\dfrac{1-2s}{s^2+4s+5}$
    (g) $\dfrac{2s-3}{s^2+2s+10}$
    (h) $\dfrac{s}{(s+1)(s^2+4)}$
    (i) $\dfrac{1}{(s+1)^2(s^2+4)}$
    (j) $\dfrac{G(s)}{s^2+1}$